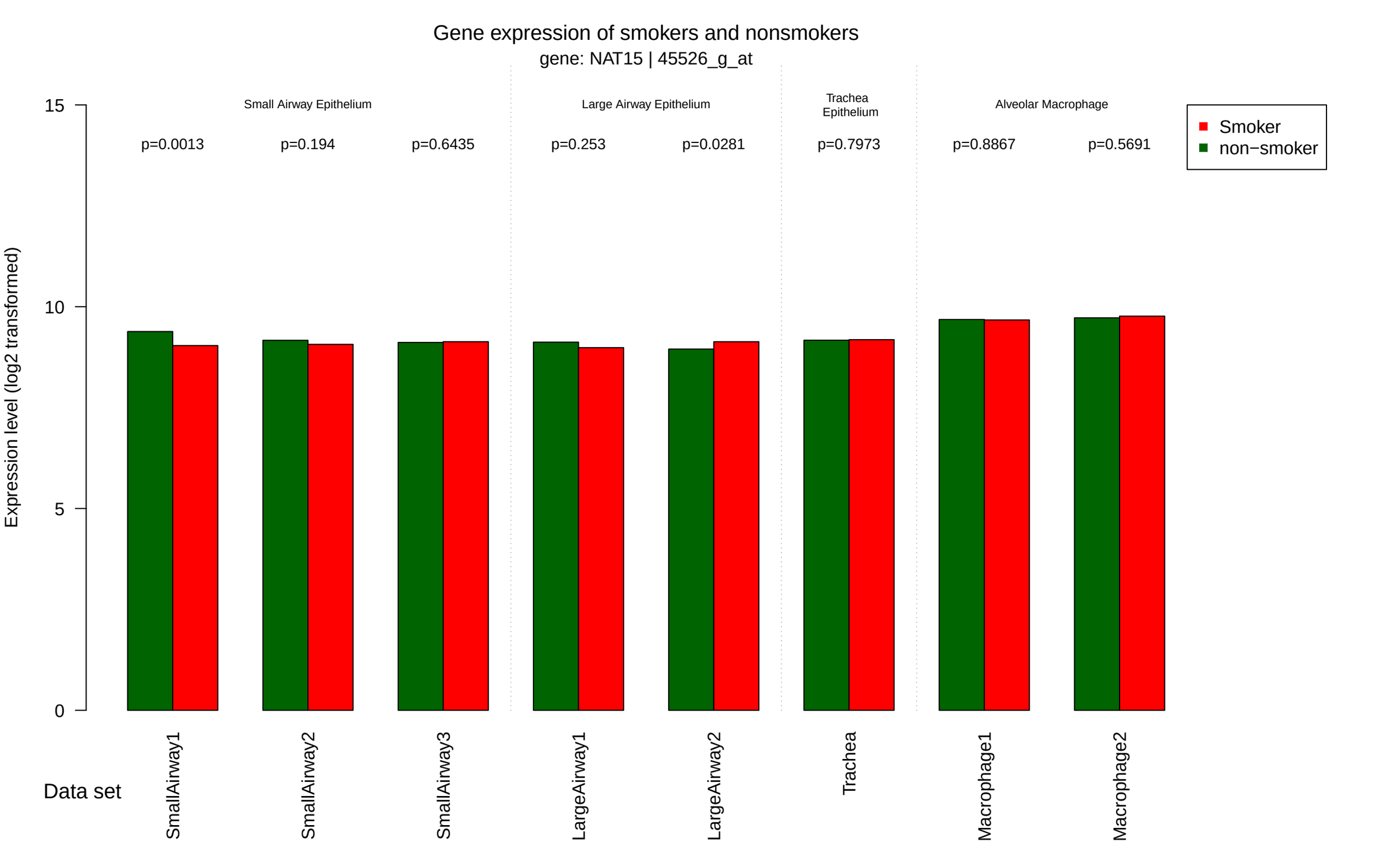
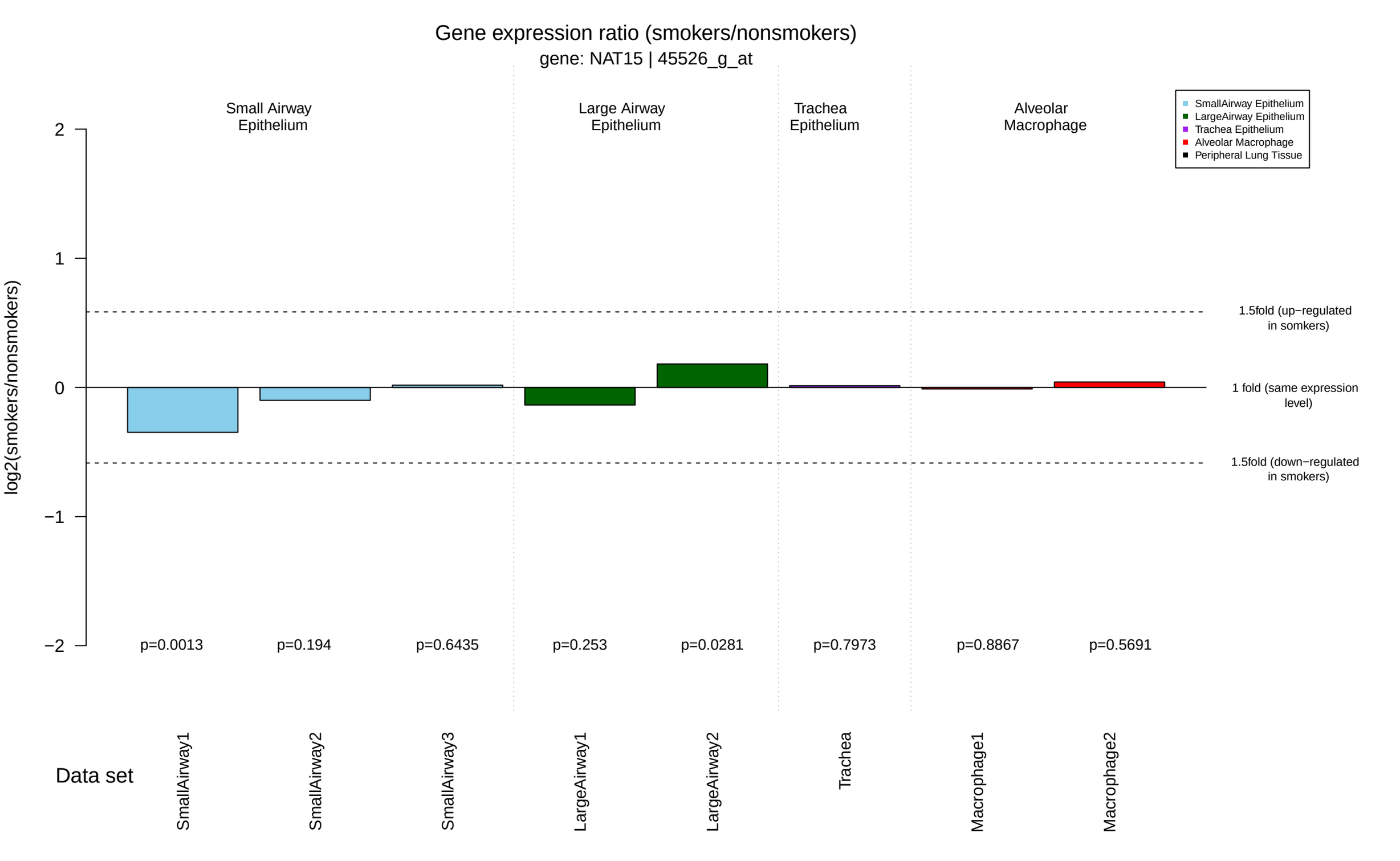
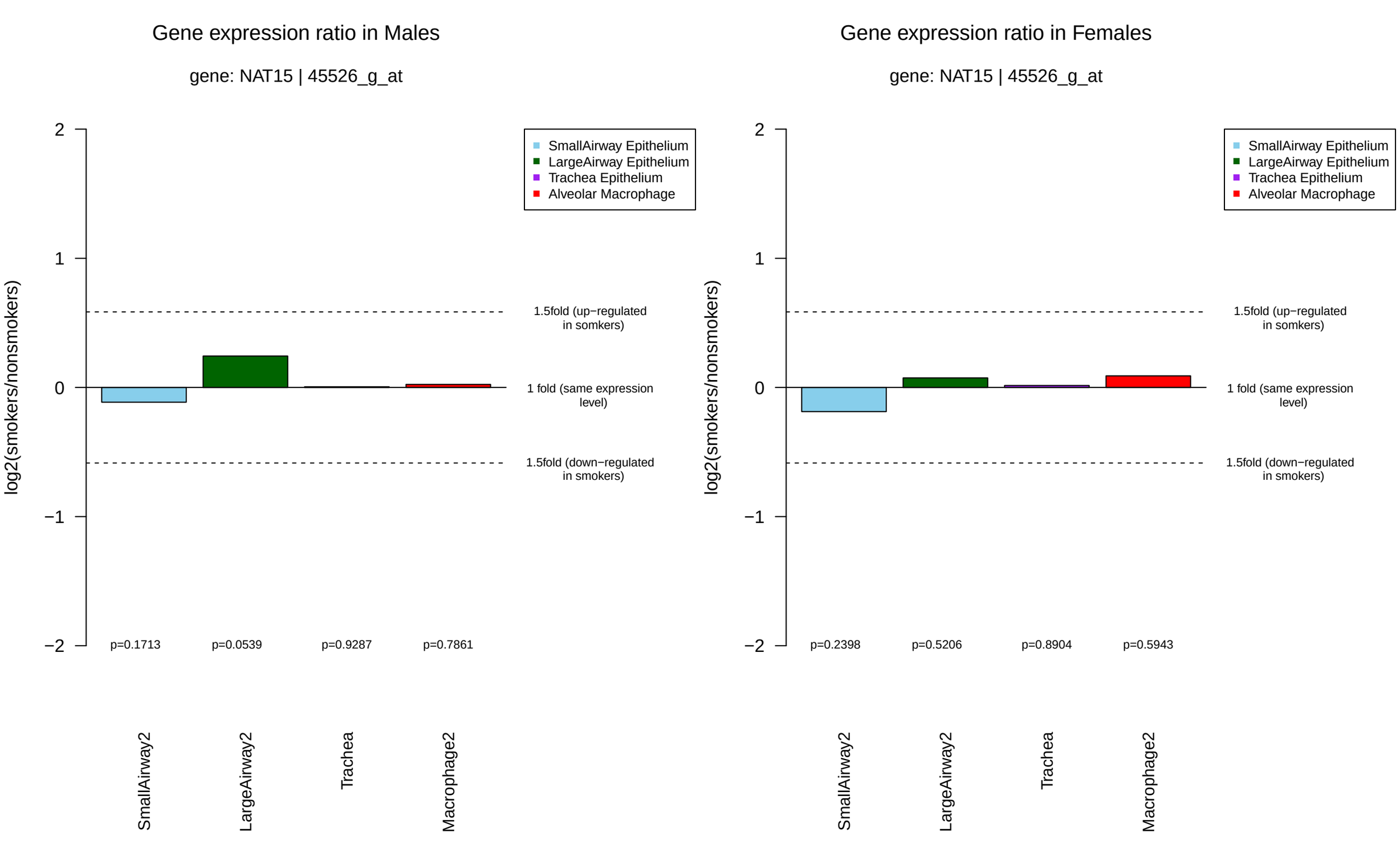
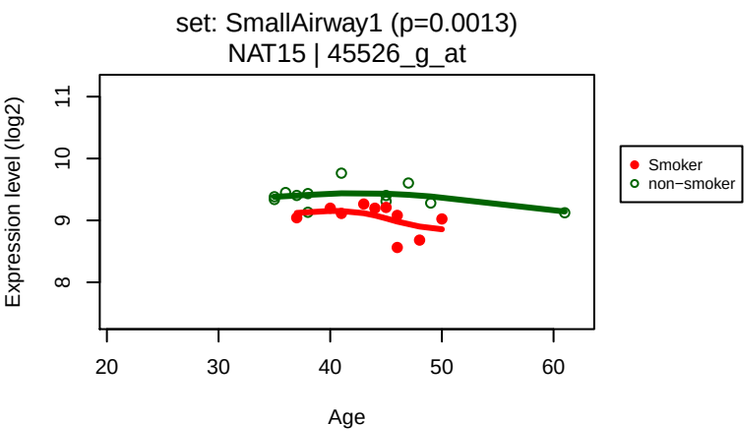
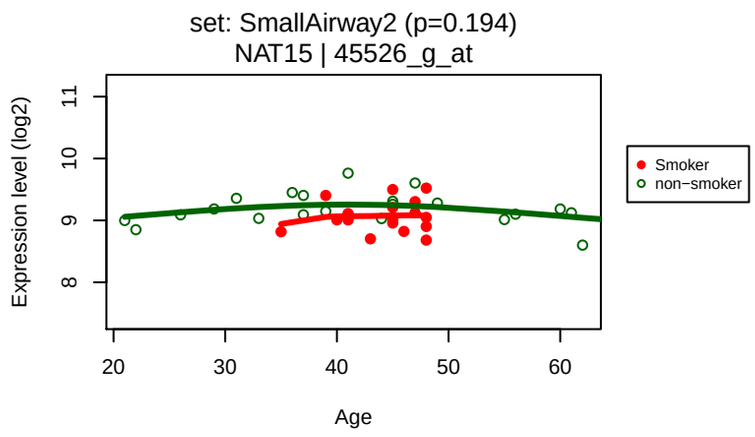
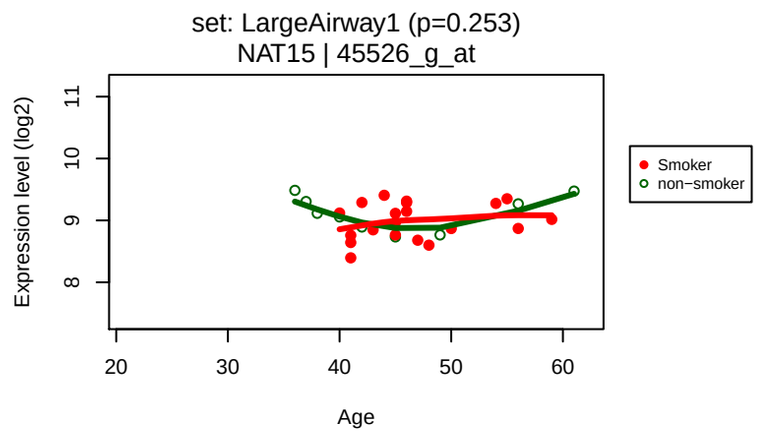

Gene expression of smokers and nonsmokers
gene: NAT15 | 45526_g_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0013
p=0.194
p=0.6435
p=0.253
p=0.0281
p=0.7973
p=0.8867
p=0.5691
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

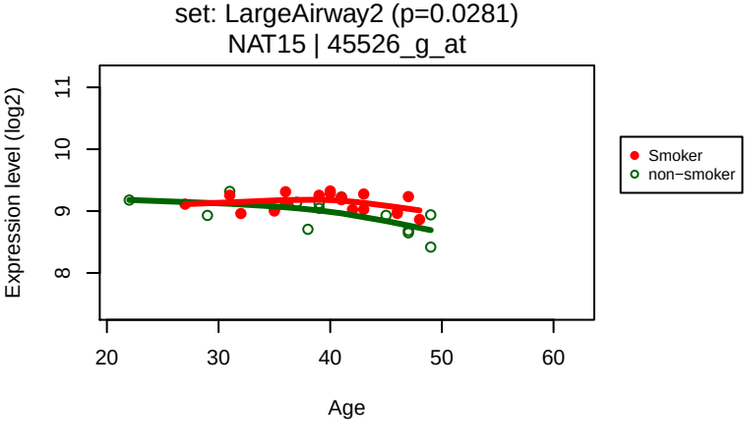

Gene expression ratio (smokers/nonsmokers)
gene: NAT15 | 45526_g_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0013
p=0.194
p=0.6435
p=0.253
p=0.0281
p=0.7973
p=0.8867
p=0.5691
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

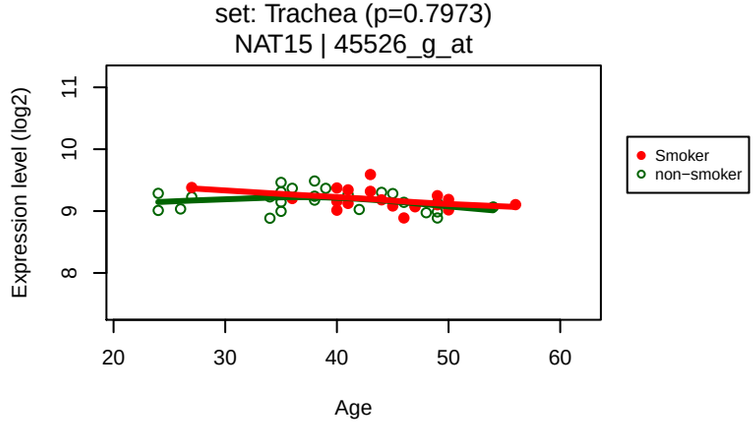

Gene expression ratio in Males
gene: NAT15 | 45526_g_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1713
p=0.0539
p=0.9287
p=0.7861
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NAT15 | 45526_g_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2398
p=0.5206
p=0.8904
p=0.5943
SmallAirway2
LargeAirway2
Trachea
Macrophage2

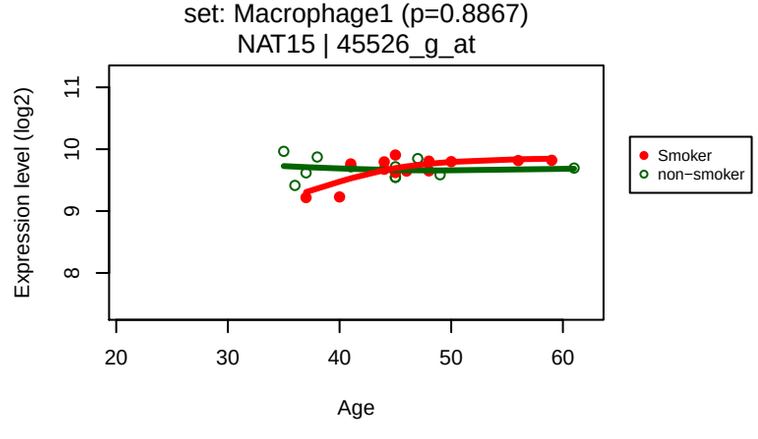

set: SmallAirway1 (p=0.0013)
NAT15 | 45526_g_at
Expression level (log2)
Age
Smoker
non−smoker

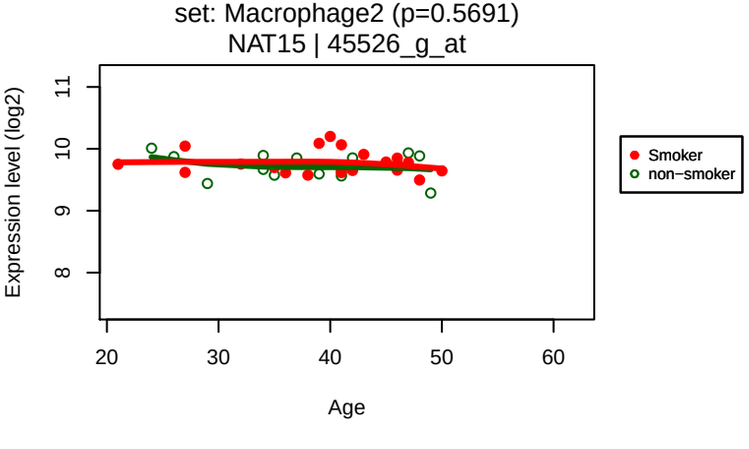

set: SmallAirway2 (p=0.194)
NAT15 | 45526_g_at
Expression level (log2)
Age
Smoker
non−smoker

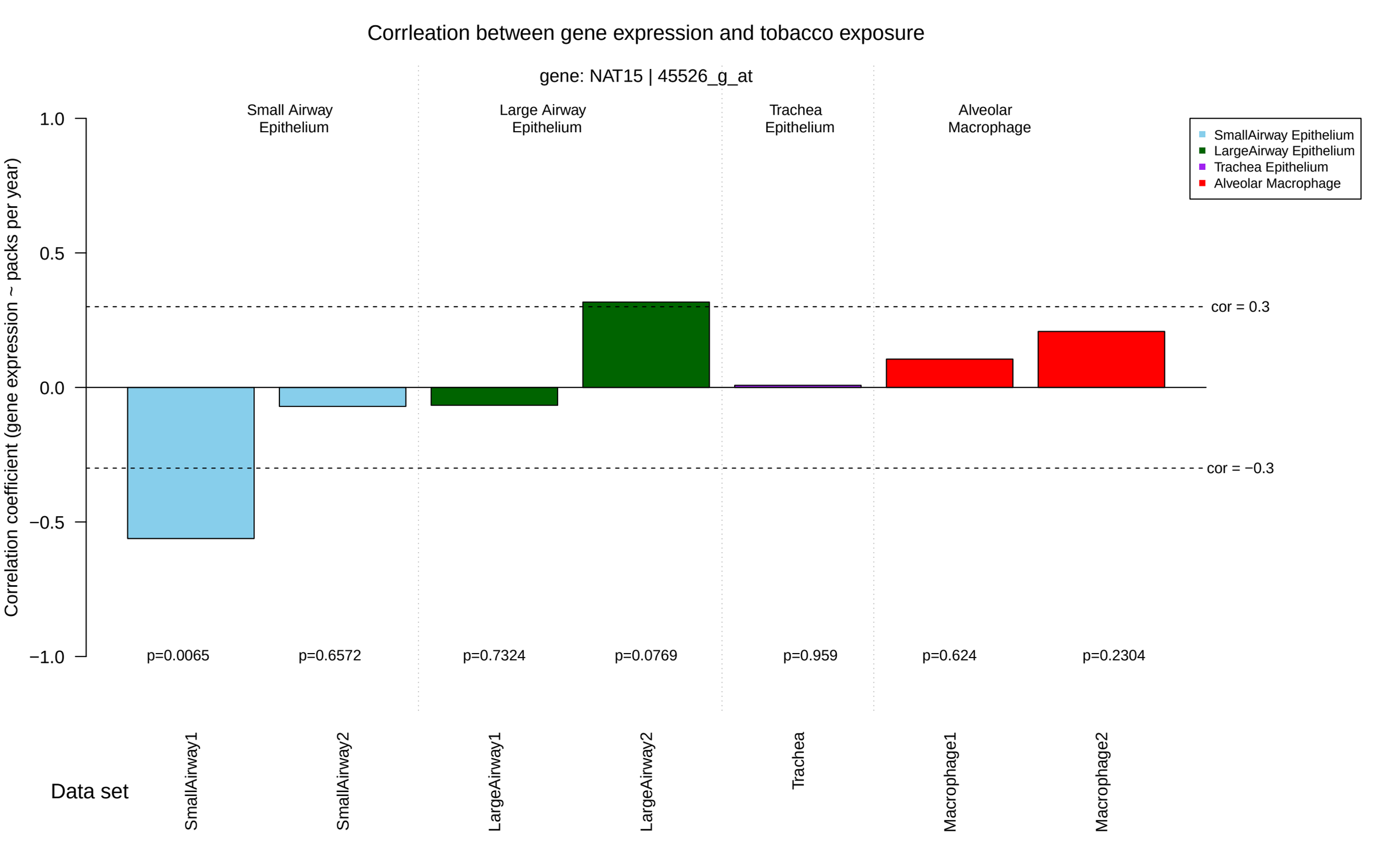

set: LargeAirway1 (p=0.253)
NAT15 | 45526_g_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0281)
NAT15 | 45526_g_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7973)
NAT15 | 45526_g_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8867)
NAT15 | 45526_g_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5691)
NAT15 | 45526_g_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NAT15 | 45526_g_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0065
p=0.6572
p=0.7324
p=0.0769
p=0.959
p=0.624
p=0.2304
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2